

Reading Ability and Semantic Processing: Isolating Speed of Access

Paul Whitney
Washington State University

George Kellas
University of Kansas

F. Richard Ferraro
Washington University

Although much attention has focused on the efficiency of access to semantic information as a determinant of reading ability, previous research has confounded information access and decision processes. The authors used a matching task to study the time course of semantic access among college students. They studied the time course of processing by varying the stimulus onset between the two members of a stimulus pair. Semantic encoding functions obtained from this paradigm were used to isolate semantic access from the decision component of the matching task. Regression analyses indicated that the often-obtained correlation between speed of semantic processing and measures of reading ability is based on unique contributions from both semantic access and decision processes. The results undermine the hypothesis that the efficiency of access to semantic codes is a relatively age-invariant contributor to individual differences in reading.

We examined speed of semantic access as a component of reading ability. Our purpose was to determine whether adult readers of varying skill levels differ in speed of access to semantic information. The question of whether semantic access is inefficient in less skilled readers has, of course, been addressed in several recent reports of evidence that poorer readers generally access semantic information more slowly (e.g., Chabot, Zehr, Prinzo, & Petros, 1984; Goldberg, Schwartz, & Stewart, 1977; Howell & Manis, 1986; Jackson & McClelland, 1979). Interest in the connection between semantic processing efficiency and reading ability has grown largely out of an interactive view of reading (e.g., Rumelhart, 1977). According to this view, stimulus-driven (bottom-up) processes and knowledge-driven (top-down) processes interact within a limited-capacity information-processing system. This implies that if lower level encoding processes are not efficient, attentional resources will be used up by coding at the word level; consequently, not enough resources will be available for higher level comprehension processes (e.g., Curtis, 1980; LaBerge & Samuels, 1974; Perfetti, 1985). In this article, we reexamine the issue of semantic processing efficiency both conceptually and empirically. Such a reexamination seems called for because the interpretation of studies on speed of access has become quite controversial. Two general concerns have been expressed about the evidence's indications that slow semantic access is a problem in less skilled readers. First, several investigators have expressed concern that semantic processing speed accounts for only a modest proportion of the variance in reading ability (e.g., Cooper & Regan, 1982; Sternberg & Powell, 1983). Correlations between measures of

processing speed and reading skill (or general verbal ability) generally fall between .30 and .40. However, it is important to bear in mind that such modest correlations are obtained with college student subjects. Because this group represents a truncated range of ability, one can expect the correlations to be higher when a wider range of ability is used. Evidence bears this out (Hunt, 1978; Sperber & McCauley, 1984). The second and perhaps more serious concern is that it is difficult to tell whether the "basic information processing tasks" used to study individual differences actually isolate semantic access (cf. Carr, 1981; Cooper & Regan, 1982; Sternberg & Powell, 1983; Whitney, 1985).

The difficulty in isolating a particular process such as semantic access stems from the fact that even so-called basic information-processing tasks may involve several subprocesses. Consequently, a thorough process analysis of relevant information-processing tasks is necessary if one is to understand information-processing differences among readers and use such data to design instructional programs.

Two important issues regarding the relation between coding efficiency and reading ability must be addressed. First, what subprocesses of word encoding are critical to individual differences? Second, is the connection between word coding subprocesses and individual differences consistent across development? There is considerable disagreement over both of these questions (cf. Stanovich, 1986).

As Perfetti (1983, 1985) pointed out, individual differences in verbal coding efficiency could arise at four levels: (a) analysis of visual input, (b) letter recognition, (c) access to the name code of a word, and (d) access to the meaning, or semantic code, of a word. With regard to developmental changes in the relative contributions of these subprocesses to individual differences, Perfetti (1985) argued that efficiency of semantic access is more likely to be a problem for younger readers. Perfetti hypothesized that name codes are so well connected to semantic codes among adults that "high-ability

This research was supported by University of Kansas General Research Fund Grant 34089-0038.

Correspondence concerning this article should be addressed to Paul Whitney, Department of Psychology, Washington State University, Pullman, Washington 99164-4830.

adults differ from low-ability adults in the basic speed of access and retrieval of this symbol name but not in whether the symbols are bound to their semantic values" (p. 169).

In brief, Perfetti's (1985) hypothesis is that the efficiency of verbal coding plays a major role in ability differences across age levels, but the specific subprocesses responsible for individual differences change with age. Differences among younger readers may be found at several levels, including speed of semantic access. Among mature readers, individual differences in verbal coding efficiency are more localized at the level of access to name codes.¹

In contrast to Perfetti's (1985) hypothesis, several studies seem to show that speed of semantic access is an important determinant of individual differences in reading even among adults (e.g., Chabot et al., 1984; Howell & Manis, 1986). If correct, then these data would indicate that the efficiency of a particular word-level subprocess, semantic access, is not only an important determinant of individual differences but is also a factor that is not developmentally limited. It thus becomes critical to determine whether the "basic information processing tasks" used to study individual differences actually isolate semantic access.

The researchers who have examined speed of semantic access have usually used some version of a categorization task (e.g., "Is a robin a bird?"); that is, subjects' speed of making decisions about category membership has been the major dependent variable of interest. Such tasks are generally acknowledged to consist of four stages: (a) perceptual encoding, (b) semantic memory access, (c) decision, and (d) response (cf. Collins & Loftus, 1975; Gitomer, Pellegrino, & Bisanz, 1983; Sperber & McCauley, 1984; Whitney, 1986).

Because several processes underlie the categorization task, it is not particularly informative to simply show an association between categorization times and reading ability. The interpretation of such a relation will depend heavily on what underlying processes are responsible for the relationship (Whitney, 1985). The general strategy behind studies in this area has been to isolate semantic access (the process most obviously related to reading) by subtracting latencies from two types of judgments that differ in the degree of semantic access required. In addition, pictures, rather than words, have often been used to show that the semantic processes investigated are general and not tied to lower level visual word encoding. Recent examples of this general strategy include the studies by Chabot et al. (1984) and by Howell and Manis (1986).

In Chabot et al.'s (1984) study, subjects were asked to decide whether a pair of words or pictures (a) were physically identical, (b) had the same name (but might differ visually in terms of upper- or lowercase), or (c) belonged to the same semantic category. Chabot et al. obtained an estimate of time to access name codes by taking the difference between reaction times (RTs) on physical identity and same-name trials. Semantic access time was estimated from the difference between same-name and same-category trials. Reading ability correlated most strongly with semantic access time, and so Chabot et al. concluded that less skilled adult readers have slower spread of activation to conceptual representations in semantic memory.

Howell and Manis (1986) used a similar logic with a slightly different task. The subjects had to verify verbal category descriptions for pictures or words. To isolate semantic processing efficiency, Howell and Manis compared verification times for superordinate descriptions ("This is an animal") with verification times for basic-level descriptions ("This is a dog"). Developmental and ability differences in performance were obtained with this processing measure. Both Chabot et al.'s (1984) study and the results obtained by Howell and Manis were taken to indicate the importance of the efficiency of semantic access in determining reading skill even among adults.

It is clear that results like those of Chabot et al. (1984) and Howell and Manis (1986) provide evidence that less skilled adult readers are deficient in *some* operation(s) involved in speeded tasks requiring semantic judgments (cf. Jackson & McClelland, 1979). However, these data are subject to several interpretations, depending on how the operations underlying the semantic task are conceptualized. Subtracting times for name judgments from times for semantic judgments, for example, unambiguously isolates speed of semantic access only if one assumes that the stages underlying matching (visual encoding, semantic access, decision, response) do not overlap in time. This assumption now seems tenuous, and several empirical studies and processing models support the notion of a continuous flow of information to the decision stage (e.g., Anderson, 1983; Grice, Nullmeyer, & Spiker, 1982; Stanovich & Pachella, 1977).

Under the assumption of overlapping stages, the latency differences between two types of judgments could be based on *speed* of access to the different codes used or on the *amount* of activation that the codes receive (see McClelland, 1979). The latter factor seems likely to influence decision processes. It is possible that the individual differences previously obtained on semantic processing tasks resulted from individual differences in decision processes rather than from such differences in speed of semantic access (Whitney, 1985). If it is processing at the decision stage that distinguishes different skill levels of readers, then the interpretation of the underlying processing deficit in poor readers could take on a very different character. Results of studies of semantic processing efficiency have been taken to indicate that lower reading ability is associated with semantic encoding difficulties at the word level. However, if it is the comparison processes needed for decision making that are slower in poor readers, then reaction times on matching tasks may correlate with reader skill because of the continuity between comparison and higher level processes (such as semantic integration) needed for between-sentence processing (Whitney, 1986).

By directly examining the time course of a particular type of judgment, one can avoid the interpretive problems associated with the assumptions underlying subtraction techniques. Several methods have been developed to trace the time course of processing (Lorch, 1982; McClelland, 1979; Posner, 1978).

¹ We do not mean to imply that this is the only difficulty that poor readers have, according to Perfetti's (1985) verbal efficiency theory. However, name code access is postulated to play a very important role.

In this study, we used a matching task in which two stimuli were presented and subjects were asked to decide whether the two stimuli belonged to the same semantic category. In our version of the matching task, we varied the time interval between the onset of the two stimuli on a trial. This time interval is known as the stimulus onset asynchrony (SOA). This task was developed by Posner and Boies (1971) and has been used to study the time course of semantic access (e.g., Lorch, 1982). One advantage of this task is that it does not require the extensive training of subjects that is required by some other procedures (see Corbett & Wickelgren, 1978). In addition, it has been used in studies of the nature of spreading activation, and thus our results can be compared with previous "normative" data on semantic memory dynamics (Lorch, 1982).

The processing model underlying the task is straightforward (see Posner, 1978). The subject begins encoding the first stimulus, which automatically primes related information in semantic memory. The second stimulus also initiates an automatic spread of activation of associated information. The decision is based on the intersection of activation of information (Anderson, 1983; Collins & Loftus, 1975; Lorch, 1982). Because RT is measured from the onset of the second stimulus, the first stimulus serves as a prime *and* as an element in the matching task. Thus, regardless of whether the stages of processing overlap in time, the longer the SOA is, the more time the subject has to encode the first stimulus and activate related semantic information before measurement of RT begins. This "head start" on activation from the first stimulus naturally results in a decrease in RT. The decrease continues with longer SOAs until sufficient information about the first stimulus has accrued so that further increases in SOA offer no more advantage. This point is known as the point of asymptote (POA) of the function. The difference between reaction times at the 0-ms SOA (no head start on processing) and reaction times at the POA (first stimulus already processed) represents the encoding time (ET) for the first stimulus. More precisely, the ET is the time needed to access enough information about the first stimulus to allow for matching (see Posner, 1978).

The RT function predicted by this processing model thus consists of two portions. There should be a linear decrease in RT over SOA until the SOA is long enough to allow full processing of the first stimulus. From this point on, the function should be flat. This function has been obtained not only with normal adults (Lorch, 1982; Sabol & DeRosa, 1976) but also with mentally retarded children (Sperber & McCauley, 1984).

In our study, a group of college students took the Nelson-Denny reading test and later performed the variable SOA matching task, for which we used pictures drawn from several taxonomic categories. We used pictures rather than words to provide an additional test of the idea that the important factor is semantic access rather than processing specific to printed words (cf. Chabot et al., 1984; Perfetti & Lesgold, 1978). It has been found consistently that the relation between speed of semantic access and reading ability is the same regardless of whether pictures or words are used as stimuli (e.g., Chabot et al., 1984). In addition, because there is strong evidence that

words and pictures access a common semantic representation (e.g., Bajo, 1988), use of pictures in our study strengthens our case for isolating semantic access apart from lower level processes specific to word recognition.

Our primary interest was in the relation between performances on the Nelson-Denny and on the ET and POA measures from the matching task. We expected that the overall variance accounted for would replicate previous research with similar subjects. Thus we expected a multiple *R* in the .30-.40 range. The critical prediction involved the significance of the increment in variance accounted for when the POA measure was added to the model. If speed of semantic access is the aspect of the matching task that is associated with reading ability, then the ET predictor should account for a significant proportion of the variance, and the POA predictor should not account for any additional variance. If the comparison and decision processes involved in matching are an important factor, then the multiple *R* should reach the expected level and account for a significant amount of variance only when the POA factor is added to the model. It is important to note that considerable previous work rules out the possibility that lower level visual processing or the simple response element in this task could be responsible for variance accounted for by the ET measure or added by the POA measure (see Jackson & McClelland, 1979; List, Keating, & Merriman, 1985).

Besides analyzing the individual subject data with regression, we also analyzed group data by using the upper and lower quartiles of ability among our subjects. We wished to determine whether the effects of SOA and category typicality obtained by Lorch (1982) in a similar task would be replicated with both high- and low-ability subjects. This provided a check for possible strategy differences in performance across ability levels.

Method

Subjects

A total of 60 undergraduates from the University of Kansas participated in partial fulfillment of an introductory psychology course requirement. All subjects were native speakers of English. Before participating in the reaction time portion of the study, all subjects were administered Form D of the Nelson-Denny reading test.² The subjects represented a very typical sample of the college-age readers for which the Nelson-Denny was designed. The mean total score was 88 (*SD* = 28). This mean is at the 53rd percentile on the Grade 14 norms for the Nelson-Denny. The scores spanned virtually the entire

² We present our results on the basis of the Nelson-Denny composite score (vocabulary and comprehension subtests combined). Virtually every reviewer of the Nelson-Denny has commented that the composite score is a good measure for differentiating among reading ability levels and has cautioned against using the subtests separately because they differ in reliability and because of the idiosyncratic nature of the paragraphs used on the comprehension subtest (see, e.g., Raygor, 1978). We merely note that the patterns of results reported throughout generally held across both subtests but were weaker for the comprehension subtest (which is the subtest lower in reliability).

range of percentile ranks, and 49% of our sample scored below the 50th percentile.

Materials and Apparatus

For the matching task, three pictures were selected from each of 10 categories: two typical exemplars and one atypical exemplar, as determined by Young and Kellas's (1973) norms. The typical exemplars had a mean rating of 1.95, and the atypical exemplars had a mean rating of 3.98 on a 7-point scale from 1 (*most typical*) to 7 (*least typical*). The names associated with typical and atypical exemplars were equated on frequency of occurrence. In addition, the pictures were selected to be as visually dissimilar as possible so that judgments would be semantic rather than visual.

One of the typical exemplars was designated as the prime for each matching trial. Each prime was paired with the relevant typical and atypical members for "same" trials and with a randomly selected typical and atypical member for "different" trials. Thus there were 40 pairs of stimuli. Five random orders of these pairs were created to form five blocks of trials. The stimuli were affixed to 15 × 23 cm white cards and displayed with an Iconix two-channel tachistoscope.

Design and Procedure

Each block was presented with a different time interval (SOA) between the onset of the first and second pictures of a trial. The SOAs used were (in ms) 0 (simultaneous), 150, 300, 450, and 600. The prime picture always appeared in the top half of the viewing field, and the target picture was displayed below the prime.

Each subject had 10 practice trials to begin, and each block was preceded by 6 additional practice trials for that SOA. The practice trials contained stimulus pictures drawn from the same set of materials as the experimental trials, but none of the same pictures were used for both types of trials. The order of blocks (hence SOAs) was randomized for each subject.

The subjects were told that they would be asked to decide whether two items belonged to the same category. A booklet indicating what pictures would be used and what categories they belonged to was then shown to subjects. Speed and accuracy of responding were stressed equally. Subjects indicated their decisions manually with telegraph-style response keys. Assignment of "same" and "different" responses to the key for the left hand or for the right hand was balanced across subjects. All reaction times were measured from the onset of the second stimulus.

Results and Discussion

The data analyses are reported in two parts. In the first set of analyses, we used data from only the subjects with the lowest and highest scores on the Nelson-Denny test. Two groups were formed from the upper and lower quartiles. The encoding functions of these groups were compared in order to get an indication of whether there were differences in how the semantic processing task was performed. The second set of analyses were the multiple regression results involving all 60 subjects in the experiment. All differences stated as being significant reached at least the .05 level.

Group Comparisons

We calculated mean RTs within each condition for each of the 30 subjects whose data were used in these analyses (see

Figure 1). Data from trials in which there was an error in responding were excluded. Trials in which the RT was greater than 2.5 standard deviations above the subject's overall mean were counted as error trials. As will be further discussed, errors were rare.

Separate $2 \times 5 \times 2$ (Group $\times$ SOA $\times$ Typicality) mixed analyses of variance (ANOVAs) were performed on same and different RTs because these two types of judgments may involve somewhat different processes (cf. Sperber & McCauley, 1984; Lorch, 1982). The analysis of data from the same judgments showed that the higher ability group had faster response times than did the lower ability group, $F(1, 28) = 6.20$, $MS_e = 283,440$; there was a significant decline in RT with SOA, $F(4, 112) = 19.31$, $MS_e = 47,262$; and judgments were faster when the target stimulus was a typical exemplar, $F(1, 28) = 22.96$, $MS_e = 3,312$. None of the interactions were significant. It is important to note that the nature of the SOA effect was as predicted: a decrease in RT over SOA until a point at which the function leveled out (see Figure 1). This result was verified with a trend analysis of the SOA effect. There were significant linear components, $F(1, 112) = 63.74$, and quadratic components, $F(1, 112) = 11.95$. No other components were significant.

The results with RTs from different judgments were parallel to the same-judgment results (see Figure 1). The high-ability group was faster overall, $F(1, 28) = 4.74$, $MS_e = 204,609$. Reaction time decreased with SOA, $F(4, 112) = 26.45$, $MS_e = 30,495$. Again, this function was composed of significant linear trends, $F(1, 112) = 85.25$, and quadratic trends, $F(1, 112) = 16.32$. Response times were faster when the target was a typical exemplar, $F(1, 28) = 37.58$, $MS_e = 4,503$. As in the same-judgment analysis, there were no significant interactions.

The important point is that the form of the SOA function was as expected for both ability groups. The marked drop in RT until the 300-ms SOA shows that subjects were performing as predicted by the accepted processing model for this task

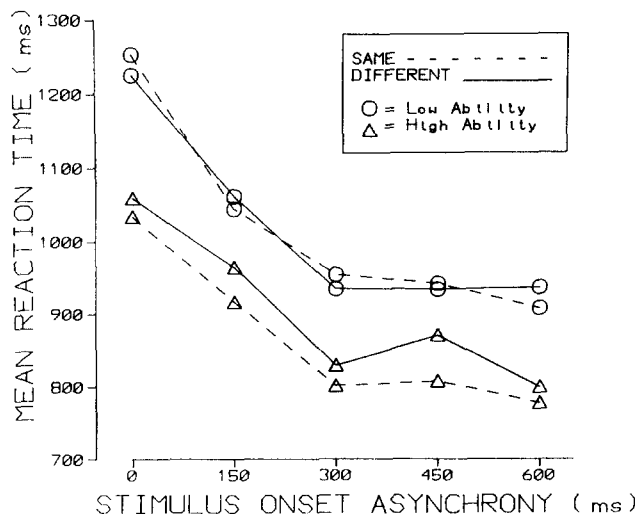


Figure 1. Mean reaction times for "same" and "different" judgments by ability and stimulus onset asynchrony.

(see Lorch, 1982; Posner, 1978). In addition, these results replicate those of Lorch (1982) in that there were no interactions of SOA and typicality. (The same result was obtained when the SOA $\times$ Typicality interaction was analyzed separately for each ability group.) Lorch interpreted this as indicating that typicality affects the strength of activation rather than rate of activation of exemplars. Although this rests on a null result, both Lorch's study and our design were sensitive enough to detect the expected effects of SOA and typicality. In any case, we can say at least that there were no indications that we were collapsing data across qualitative differences when we examined quantitative differences in performance. In no other study in this area have researchers made such an assessment.

We also examined error rates on the same and different trials. The error rates were low (less than 10% in all conditions), and they were relatively uniform. No speed-accuracy trade-offs were evident. In the ANOVAs of error rates, the only significant effect was one of typicality on same-judgment trials, $F(1, 28) = 6.83$, $MS_e = 4.39$. There were more errors in atypical trials (8.0%) than in typical trials (6.1%). No other effects approached significance, and both ability groups displayed similar levels of accuracy of performance. Across same and different judgments, the high-ability group had a mean error rate of 5.7%, and the comparable rate for the low-ability group was 6.6%.

To summarize the group comparisons: We replicated prior results by finding ability differences in categorization times. In fact, the overall mean differences between the ability groups (153 ms on same judgments and 113 ms on different judgments) were very similar to the group differences obtained in prior research (e.g., Chabot et al., 1984; Howell & Manis, 1986). We are also reasonably confident that these group differences do not reflect qualitative differences in how the task was performed. However, unlike prior researchers, we would not conclude from these results that the ability differences necessarily lie in speed of semantic access. That hypothesis was tested in the regression analyses reported next.

Regression Analyses

Estimates of ETs and POAs were obtained for each subject. These were calculated separately for same and different judgments. On the basis of the group results, we used the average of each subject's reaction time at the longest two SOAs as the POA. The ET measure was the difference between the reaction time at the 0-ms SOA and the POA.

The results of the regression are shown in Table 1. ET was forced into the model first because our interest was in the variance contributed by POA when ET was already accounted for. The results from the same judgments clearly support the hypothesis that there was a significant increment in variance accounted for when the POA measure was added to the model. Besides the significant F -to-enter for the POA factor, the R value was significant, and fell within the anticipated range, only when the POA component was added to the model, $F(2, 57) = 3.66$, $p < .04$.

The effects were not as strong when different judgments were used, but the pattern paralleled the same-judgment results. The increment in variance accounted for when the POA component was added did not quite reach significance, but its overall contribution was about equal to that of the ET component. In addition, the R value was marginally significant when the POA component was added, $F(2, 57) = 2.91$, $p = .06$.

It is not unusual to find a weaker relation between ability and RTs derived from different judgments. This was reported by Howell and Manis (1986) and can also be seen in Chabot et al.'s (1984) data.

Thus our overall results based on the full regression models precisely replicated those of prior studies on the relation between categorization times and reading ability. Yet we have shown that the basis for this relation is more complex than previously thought. Although semantic access time had some importance (it had the highest simple correlation with reading performance), it was not solely responsible for the relation between reading ability and the semantic categorization task.

Two further analyses strengthen our interpretations. First, the two predictors in the regression model (ET and POA) were not highly correlated enough to yield the interpretive problems associated with multicollinearity (see Pedhazur, 1982). For the same judgments, the correlation was $-.32$, and for different judgments, it was $-.22$. Second, one analysis across same and different judgments made sense in our context. As a manipulation check, we calculated the average ETs for the two types of judgments. If the ET measure is isolating semantic access, then approximately the same value should be obtained for both judgments because the primes were identical in each case. The values were 325 ms for the same data and 305 ms for the different data. This difference did not approach significance ($p > .20$).

Also, the absolute magnitude of our estimate of encoding time was very consistent with estimates based on other paradigms. A wide variety of techniques have converged on esti-

Table 1
Summary of Regression Analyses

Judgment/step	Predictor	R	R^2	R^2 change	F -to-enter
Same					
1	ET	.24	.06	.06	3.41*
2	POA	.34	.12	.06	3.75**
Different					
1	ET	.22	.05	.05	2.84*
2	POA	.31	.10	.05	2.90*

Note. ET = encoding time; POA = point of asymptote.

*.09 $> p > .05$. ** $p < .05$.

mates of 200–400 ms as the time needed for sufficient information about simple pictures or words to accrue in order to begin to make semantic judgments. The techniques used include not only reaction time methods (e.g., Posner, 1978) but also methods based on event-related potentials (e.g., Hillyard & Kutas, 1983). The average semantic access times that we obtained seem to be good estimates in light of recent interpretations of the P-300 component of event-related potentials (Van Petten & Kutas, 1987).

Conclusions and Implications

The major conclusion that can be drawn from this study is that previous investigations of the relation between reading ability and semantic processing efficiency should be interpreted with caution. Because they assumed that the locus of their effects was solely in the access to semantic codes, several previous investigators have interpreted their results as indicative of individual differences in word-level processing. However, our results indicate that factors other than semantic access speed are involved in the relation between matching tasks and reading ability. On the basis of these results, it is conceivable that the relation is not due to word-level processing alone. The comparison processes needed for integrating successive elements of text could be correlated with the comparison processes used in the matching task (see Whitney, 1985).

Our data thus considerably weaken the argument that efficiency of access to semantic codes represents an age-invariant contributor to individual differences in reading ability. Although speed of semantic access may be an important determinant of reading skill among children (e.g., Perfetti, 1985), it played only a weak role among our adult subjects. It seems likely that a much larger role is played by individual differences in the comprehension skills that facilitate information integration (see Corely, 1988). This suggests a hypothesis that deserves further study. It may be that a modest relation between the comparison processes of the categorization task and the integration processes of comprehension resulted in the contribution of the POA component to the variance accounted for.

By showing that the relation between "basic" measures of information processing and more global measures of ability are not always as simple as often hypothesized, our study highlights two important methodological points. First, to relate speed of some basic process to more global measures of ability, it is essential to study the time course of processing. The variable SOA paradigm used here is well suited for this purpose. Second, investigators should look for qualitative differences in performance between groups before drawing strong conclusions about the importance of quantitative differences. Even though no qualitative differences were found among subjects in our study, it is quite possible that such differences could be obtained when wider ranges of ability (or different age groups) are used (see Whitney, 1985). As several investigators have pointed out, it is possible to find strategy differences that affect performance even on tasks that appear to be very simple or automatic (e.g., Baron, 1978; Cooper & Regan, 1982; Siegler, 1987).

References

- Anderson, J. R. (1983). A spreading activation theory of memory. *Journal of Verbal Learning and Verbal Behavior*, 22, 261–295.
- Bajo, M. T. (1988). Semantic facilitation with pictures and words. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 14, 579–589.
- Baron, J. (1978). Intelligence and general strategies. In G. Underwood (Ed.), *Strategies of information processing* (pp. 403–450). New York: Academic Press.
- Carr, T. H. (1981). Building theories of reading ability: On the relation between individual differences in cognitive skills and reading comprehension. *Cognition*, 9, 73–114.
- Chabot, R. J., Zehr, H. D., Prinz, O. V., & Petros, T. V. (1984). The speed of word recognition subprocesses and reading achievement in college students. *Reading Research Quarterly*, 19, 147–161.
- Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review*, 82, 407–428.
- Cooper, L. A., & Regan, D. T. (1982). Attention, perception, and intelligence. In R. J. Sternberg (Ed.), *Handbook of human intelligence* (pp. 123–169). New York: Cambridge University Press.
- Corbett, A. T., & Wickelgren, W. A. (1978). Semantic memory retrieval: Analysis by speed accuracy tradeoff functions. *Quarterly Journal of Experimental Psychology*, 30, 1–15.
- Corely, P. J. (1988). A developmental analysis of sentence comprehension abilities in good and poor readers. *Educational Psychologist*, 23, 57–75.
- Curtis, M. E. (1980). Development of components of reading skill. *Journal of Educational Psychology*, 72, 656–669.
- Gitomer, D. H., Pellegrino, J. W., & Bisanz, J. (1983). Developmental change and invariance in semantic processing. *Journal of Experimental Child Psychology*, 35, 56–80.
- Goldberg, R. A., Schwartz, S., & Stewart, M. (1977). Individual differences in cognitive processes. *Journal of Educational Psychology*, 69, 9–14.
- Grice, G. R., Nullmeyer, R., & Spiker, V. A. (1982). Human reaction time: Toward a general theory. *Journal of Experimental Psychology: General*, 111, 135–153.
- Hillyard, S. A., & Kutas, M. (1983). Electrophysiology of cognitive processing. *Annual Review of Psychology*, 34, 33–61.
- Howell, M. J., & Manis, F. R. (1986). Developmental and reader ability differences in semantic processing efficiency. *Journal of Educational Psychology*, 78, 124–129.
- Hunt, E. (1978). Mechanics of verbal ability. *Psychological Review*, 85, 109–130.
- Jackson, M. D., & McClelland, J. L. (1979). Processing determinants of reading speed. *Journal of Experimental Psychology: General*, 108, 151–181.
- LaBerge, D., & Samuels, S. J. (1974). Toward a theory of automatic information processing in reading. *Cognitive Psychology*, 6, 293–323.
- List, J. A., Keating, D. P., & Merriman, W. E. (1985). Differences in memory retrieval: A construct validity investigation. *Child Development*, 65, 138–151.
- Lorch, R. F. (1982). Priming and search processes in semantic memory: A test of three models of spreading activation. *Journal of Verbal Learning and Verbal Behavior*, 21, 468–492.
- McClelland, J. L. (1979). On the time relations of mental processes: An examination of systems of processes in cascade. *Psychological Review*, 87, 329–354.
- Pedhazur, E. J. (1982). *Multiple regression in behavioral research* (2nd ed.). New York: Holt, Rinehart & Winston.
- Perfetti, C. A. (1983). Individual differences in verbal processes. In R. F. Dillon & R. R. Schmeck (Eds.), *Individual differences in*

- cognition* (Vol. 1, pp. 65-104). New York: Academic Press.
- Perfetti, C. A. (1985). *Reading ability*. New York: Oxford University Press.
- Perfetti, C. A., & Lesgold, A. (1978). Discourse comprehension and sources of individual differences. In M. Just & P. Carpenter (Eds.), *Cognitive processes in comprehension* (pp. 141-182). Hillsdale, NJ: Erlbaum.
- Posner, M. I. (1978). *Chronometric explorations of mind*. Hillsdale, NJ: Erlbaum.
- Posner, M. I., & Boies, S. J. (1971). Components of attention. *Psychological Review*, 78, 391-408.
- Raygor, A. L. (1978). Review of the Nelson-Denny Reading Test. In O. K. Buros (Ed.), *The eighth mental measurements yearbook*. (Vol. 2, pp. 735-736). Highland Park, NJ: Gryphon.
- Rumelhart, D. E. (1977). Toward an interactive model of reading. In S. Dornic (Ed.), *Attention and performance VI* (pp. 265-303). Hillsdale, NJ: Erlbaum.
- Sabol, M. A., & DeRosa, D. V. (1976). Semantic encoding of isolated words. *Journal of Experimental Psychology: Human Learning and Memory*, 2, 58-68.
- Siegler, R. A. (1987). The perils of averaging data over strategies: An example from children's addition. *Journal of Experimental Psychology: General*, 116, 250-264.
- Sperber, R., & McCauley, C. (1984). Semantic processing efficiency in the mentally retarded. In P. Brooks, R. Sperber, & C. McCauley (Eds.), *Learning and cognition in the mentally retarded* (pp. 141-163). Hillsdale, NJ: Erlbaum.
- Stanovich, K. E. (1986). Matthew effects in reading: Some consequences of individual differences in acquisition of literacy. *Reading Research Quarterly*, 21, 360-406.
- Stanovich, K. E., & Pachella, R. G. (1977). Encoding, stimulus-response compatibility, and stages of processing. *Journal of Experimental Psychology: Human Perception and Performance*, 3, 411-421.
- Sternberg, R. S., & Powell, J. S. (1983). Comprehending verbal comprehension. *American Psychologist*, 38, 878-893.
- Van Petten, C., & Kutas, M. (1987). Ambiguous words in context: An event-related potential analysis of the time-course of meaning activation. *Journal of Memory and Language*, 26, 188-208.
- Whitney, P. (1985). Reading achievement and isolable subprocesses. *Reading Research Quarterly*, 20, 654-657.
- Whitney, P. (1986). Developmental trends in speed of semantic memory retrieval. *Developmental Review*, 6, 57-79.
- Young, M., & Kellas, G. (1973). *Typicality norms for pictures*. Unpublished manuscript, University of Kansas.

Received April 13, 1989

Revision received January 2, 1990

Accepted February 19, 1990 ■